

Cervical Shortening 07. Interventions Compared

Companion reading handout for simultaneous listening.

Audio course on cervical shortening during pregnancy. Episode seven. Interventions compared.

Opening frame

This episode is where we put the tools on the table.

Vaginal progesterone. Cerclage. Pessary. 17-OHPC. Bed rest. Tocolysis. Steroids. Magnesium.

The mistake is to think of these as a menu for short cervix.

They are not.

Each tool belongs to a different phenotype, a different mechanism, or a different time horizon.

Dominant model

The dominant model is this.

Intervention choice depends on the patient phenotype, not the cervical length alone.

Evidence and mechanisms

Let us start with vaginal progesterone.

Vaginal progesterone has the cleanest evidence for asymptomatic singleton short cervix before twenty four weeks, especially cervical length less than or equal to twenty millimeters. SMFM recommends vaginal progesterone for that group. For twenty one to twenty five millimeters, SMFM recommends considering it through shared decision-making.

Why does it make sense?

Progesterone helps maintain uterine quiescence and opposes cervical ripening. In human labor, the issue is not simply low serum progesterone. It may be local functional progesterone withdrawal, receptor changes, and inflammatory signaling. Vaginal progesterone may act near the cervix and lower uterine segment, which makes it biologically plausible for short cervix.

But do not overextend the mechanism.

If the cervix is shortening because of advanced infection, PPRM, significant dilation, bleeding, or uterine overdistension, progesterone may not reverse the pathway. Mechanism gives plausibility. It does not erase phenotype.

The studied regimens matter. Fonseca used two hundred milligrams of micronized vaginal progesterone nightly in women with very short cervix. Hassan's PREGNANT trial used ninety milligrams of vaginal progesterone gel in cervical length ten to twenty millimeters. The Romero individual patient data meta-analysis supported reduction in early preterm birth and neonatal morbidity outcomes in asymptomatic singleton short cervix.

SMFM notes that ninety milligram gel and two hundred milligram micronized capsules are the most studied formulations, but evidence is insufficient to recommend one specific dose or formulation as universally superior.

The resident reflex: vaginal progesterone is a short-cervix intervention in selected patients. It is not a generic cure for all preterm birth risk.

17-OHPC

Now 17-OHPC.

This is the intervention residents must unlearn carefully.

17-alpha hydroxyprogesterone caproate is not vaginal progesterone. It is not recommended for treatment of short cervix. SMFM recommends against 17-OHPC, including compounded formulations, for short cervix. ACOG's updated progestogen advisory states that intramuscular 17-OHPC is not recommended for primary prevention of recurrent spontaneous preterm birth.

The teaching line is simple:

Do not substitute 17-OHPC for vaginal progesterone because both contain the word progesterone.

Cerclage

Now cerclage.

Cerclage is powerful when the indication is right and misleading when the indication is wrong.

The three categories matter.

History-indicated cerclage is based on a compelling obstetric history, such as recurrent painless midtrimester losses or prior exam-indicated cerclage.

Ultrasound-indicated cerclage is based on a high-risk history plus current cervical shortening, usually prior spontaneous preterm birth and cervical length below twenty five millimeters before twenty four weeks.

Physical-exam-indicated cerclage is based on current painless cervical dilation, often with exposed membranes, after excluding labor, PPROM, infection, bleeding, and fetal conditions that change goals of care.

If you cannot name the cerclage category, you are not ready to recommend cerclage.

Now evidence.

The To trial in broadly screened women with cervical length less than or equal to fifteen millimeters did not show a broad benefit from cerclage. That supports SMFM's recommendation against routine cerclage for a no-prior-spontaneous-preterm-birth singleton with cervical length ten to twenty five millimeters and no dilation.

Owen two thousand nine is different. It studied women with prior spontaneous preterm birth before thirty four weeks whose current cervix shortened to less than twenty five millimeters. Overall benefit was not uniform, but previable birth and perinatal mortality were lower, and benefit appeared concentrated in the very short subgroup, especially less than fifteen millimeters.

FIGO recommends ultrasound-indicated cerclage for cervical length below twenty five millimeters in women with at least one prior spontaneous preterm birth or midtrimester loss. NICE offers a choice of prophylactic vaginal progesterone or cerclage when both prior spontaneous preterm birth or loss from sixteen to thirty four weeks and cervical length twenty five millimeters or less are present.

So the clinical synthesis is this:

Cerclage is not for every short cervix. It is for selected histories, selected ultrasound phenotypes, and selected dilation presentations.

Cervical pessary

Now pessary.

Pessary teaches trial humility.

The mechanism is attractive. A pessary might change the uterocervical angle, redistribute pressure, reduce opening of the internal os, or protect the mucus plug. It is nonoperative. It feels reasonable.

Goya's PECEP trial was strongly positive in singleton short cervix, with a large reduction in spontaneous delivery before thirty four weeks. But Nicolaides' large trial did not replicate a benefit. SMFM recommends against routine cervical pessary for singleton short cervix prevention. In multiples, ProTWIN did not show overall benefit, though subgroup signals appeared.

SuPPoRT two thousand twenty four compared cerclage, pessary, and vaginal progesterone in singleton pregnancies with cervical length less than twenty five millimeters, mostly with risk factors. Preterm birth before thirty seven weeks was not significantly different among groups: about thirty point five percent with cerclage, thirty one point two percent with pessary, and twenty four point two percent with progesterone. This suggests no clear superiority in that trial population, but it does not collapse all guideline distinctions. It was not a license to ignore history, dilation, twins, or local practice.

The resident reflex: plausible device plus one positive trial is not enough for routine practice when later evidence is inconsistent.

Bed rest and activity restriction

Now bed rest.

Bed rest is an intervention too, even though it is often prescribed casually. It has harms and no proven prevention role for isolated cervical shortening. SMFM Consult number fifty recommends against routine activity restriction for shortened cervix or preterm birth risk. ACOG Committee Opinion eight zero four says activity restriction should not be prescribed routinely to reduce preterm birth. AWARE two thousand twenty six, using accelerometers in patients with short cervical length, found no latency benefit from sedentary activity, and fewer than three thousand five hundred steps per day was associated with earlier birth and higher preterm birth before thirty four weeks.

So do not say, it cannot hurt.

It can hurt.

Now tocolysis, steroids, and magnesium.

These are often mentioned in the same conversation because the cervix is short and everyone is worried. But they are not primary treatments for asymptomatic short cervix.

Tocolysis is a short-term tool for threatened or established preterm labor, usually to gain time for corticosteroids or transfer. It is not long-term prevention for a closed short cervix.

Antenatal corticosteroids are used when preterm birth risk is sufficiently high in an appropriate gestational window. A short cervix alone does not always mean delivery is imminent enough to justify steroids. The decision depends on gestational age and short-term delivery risk.

Magnesium sulfate is neuroprotection when early preterm delivery is anticipated, often before thirty two weeks depending on protocol. It is not a treatment for cervical length.

GBS prophylaxis belongs to threatened preterm birth management according to local protocol.

Antibiotics are not indicated for preterm labor with intact membranes unless there is another infection indication. PPROM is different.

Bedside algorithm

Now let us build a tool-by-phenotype map.

No-prior-spontaneous-preterm-birth singleton, closed cervix, less than or equal to twenty millimeters before twenty four weeks: vaginal progesterone.

No-prior singleton, twenty one to twenty five millimeters: consider vaginal progesterone through shared decision-making.

No-prior singleton, ten to twenty five millimeters, no dilation: do not place cerclage routinely.

Prior spontaneous preterm birth plus current cervical length below twenty five millimeters: discuss progesterone and ultrasound-indicated cerclage, especially if the cervix is very short and the prior birth was early.

Classic recurrent painless midtrimester losses: think history-indicated cerclage.

Painless dilation with intact membranes and no infection, no labor, no bleeding: consider physical-exam-indicated cerclage in the right setting.

Twins with short cervix: use cervical length for risk stratification; do not routinely use progesterone, pessary, or cerclage outside trials according to SMFM.

Short cervix with contractions: triage for preterm labor.

Short cervix with leakage: evaluate for PPROM.

Short cervix with bleeding: do not treat as uncomplicated.

Now a case.

Clinical cases

Twenty one weeks, singleton, no prior preterm birth, cervix eighteen millimeters, closed.

Best-supported tool: vaginal progesterone. Not cerclage. Not pessary. Not 17-OHPC. Not bed rest.

Second case.

Twenty one weeks, prior spontaneous preterm birth at twenty nine weeks, cervix fourteen millimeters.

Now cerclage is a serious option, and vaginal progesterone strategy is also relevant. The prior history and very short cervix change the tool selection.

Third case.

Twenty two weeks, twins, cervix eighteen millimeters, closed.

High risk. But SMFM does not recommend routine progesterone, pessary, or cerclage outside a trial. Use MFM judgment and surveillance, not singleton reflexes.

Fourth case.

Twenty two weeks, three centimeters dilated, membranes visible.

Do not reach first for progesterone. This is possible exam-indicated cerclage after excluding labor, PPROM, infection, bleeding, and fetal conditions.

Final synthesis

Let us close with the resident reflex.

Do not ask, what do we do for a short cervix?

Ask, which short cervix is this?

The right intervention is not chosen by the millimeter alone. It is chosen by phenotype: history, gestational age, plurality, symptoms, dilation, membranes, infection, bleeding, and the treatment window.